Heer Kumari Perchani

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Authorized to work in Canada | Available for a 4–16 month internship starting January 2027

Education

University of Alberta Expected Graduation: 2028
B.Sc. in Computing Science, Minor in Statistics *Edmonton, AB*
Co-op: Science Internship Program (SIP), Faculty of Science
Awards: Dean's Honour Roll, First Class Standing
Coursework: Machine Learning, Reinforcement Learning, Natural Language Processing, Computer Vision, Probability & Statistics

Technical Skills

Languages: Python, C++, SQL, R, JavaScript, Bash
ML & AI: PyTorch, TensorFlow, Scikit-Learn, Hugging Face, vLLM, DeepSpeed, OpenCV, LLMs, Reinforcement Learning (PPO), CNNs, Transfer Learning, Random Forests, Gemini/OpenAI APIs
Data: Pandas, NumPy, Matplotlib, Seaborn, BigQuery, DuckDB, SQL Server
Tools & Infrastructure: Linux, Docker, Multi-GPU Training, Slurm/HPC, Weights & Biases, Conda, Git/GitHub, Jupyter, Google Colab, Flask

Experience

Undergraduate Researcher Sep 2026 – Present
Computing Research Association (UR2PHD Program) *Edmonton, AB*
- Reproducing SPIRAL (ICLR 2026), a multi-agent self-play reinforcement learning (RL) framework that improves LLM reasoning by training on zero-sum games
- Set up the released training stack (PyTorch, vLLM, DeepSpeed, Hugging Face) on a shared 8-GPU Linux server and verified the full RL loop (game rollouts, PPO policy update, weight sync) with DeepSeek-R1-Distill-Qwen-1.5B on 4 GPUs
- Debugged dependency and launch-config failures (flash-attn build, missing environment wrappers, API-gated evaluation); now porting the pipeline to the Narval HPC cluster (Slurm) to scale to Qwen3-4B

Machine Learning Fellow May 2026 – Jun 2026
AI4Good Lab & Amii *Edmonton, AB*
- Co-developed FurCast, an ML tool predicting shelter animals' length-of-stay and outcome; reached 86.6% accuracy on dog length-of-stay vs. a 60% baseline
- Cleaned and merged 8 datasets (80,000+ records) with municipal, economic and demographic data, using the Gemini API to fill missing breed attributes for feature engineering
- Trained per-species Random Forest models with hyperparameter sweeps and a log-loss-optimized ensemble, guided by input from Humane Canada and 5 shelters
- Trained ResNeXt-50 and EfficientNetV2 CNNs for cassava leaf disease classification (Kaggle), reaching 87.38% accuracy

Teaching Assistant – CMPUT 175 Sep 2025 – Dec 2025
University of Alberta *Edmonton, AB*
- Led weekly Python labs for 130+ students on data structures and algorithms; graded 20+ assignments a week with feedback on code structure and logic

Software & Data Analyst Intern Jun 2025 – Aug 2025
SoftDev *Remote*
- Wrote and tested SQL Server queries and parameterized reports for 5 ERP finance modules (billing, general ledger, vouchers, trial balance, ad scheduling)

Projects

Emotion Recognition Web App | *PyTorch, Flask, JavaScript* GitHub
- Designed and trained a 4-block CNN from scratch in PyTorch to classify faces into 5 emotions, reaching 71% accuracy
- Deployed the model behind a Flask backend with an image-preprocessing pipeline and a JavaScript upload interface for real-time inference

Equity Factor Research & Backtesting | *SQL (DuckDB), Python* GitHub
- Built a research pipeline on 619K daily S&P 500 prices with 11 automated checks for look-ahead bias and data integrity; tested 5 factors via rank IC and a cost-aware long-short backtest

Head Movement Navigation | *JavaScript, FaceAPI.js* GitHub
- Won the DivE category at HackED 2025 with a computer-vision Chrome extension that maps head movements to scrolling and page navigation